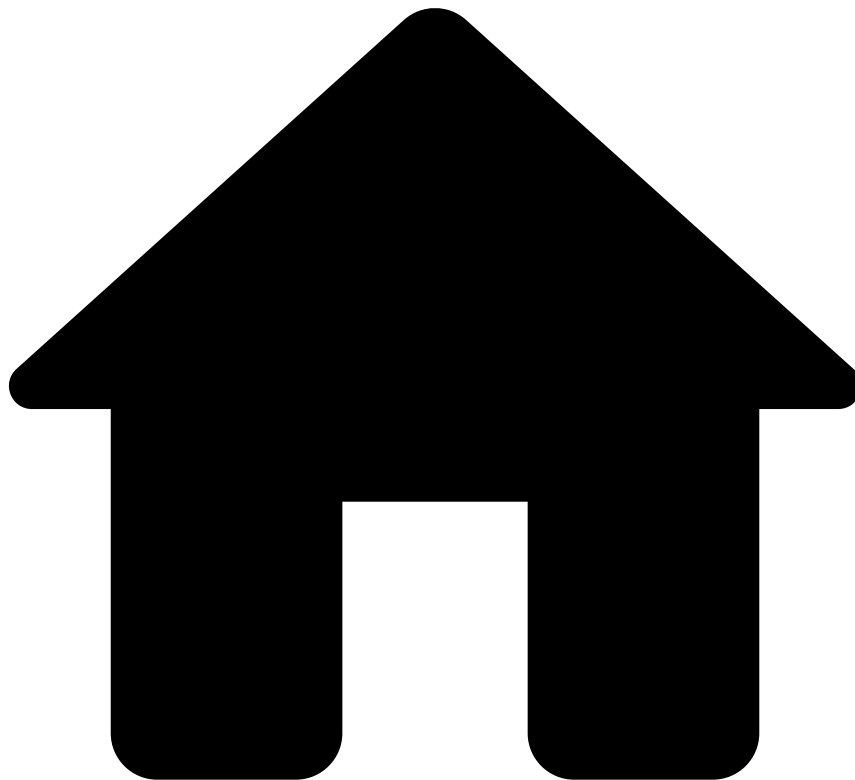


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- [Core Concepts](#)
- Models

On this page

Models

Was this page helpful? 

A **model** represents knowledge collected about the behavior of a **target variable** in a data source. You collect this knowledge by running a machine learning algorithm against a historical dataset, to investigate the data patterns in it. This process is called [training a model](#). You create a model in order to use it later in a [workflow](#) to predict the value of the target variable in live data.

After training a model, you can evaluate its accuracy and compare it to other models. For this, Pulse generates statistics in a so-called [model evaluation](#).

Models and model evaluations are organized into **model projects**.

Model life cycle

Models give your application an intelligent edge by allowing you to configure, train, fine tune and deploy a binary classifier which will score each instance. The model lifecycle is fully aligned with Pulse's [data science loop](#).

Before using a model to score a system's incoming transactions for fraud detection, it must be properly trained using representative and verified data labeled for your target variable, for example, both fraud and non-fraud cases.

Model usage is separated into two distinct lifecycle phases:

- **Training and evaluation** (in the data science environment)

Is an offline process that can be repeated with different settings and data until you are satisfied with the results.

The performance of trained models is [evaluated](#) using common performance metrics, such as F1-Measure, False Positive Rate, among others and by visualizing and comparing its ROC curve.

- **Scoring** (in the production environment)

When model performance is good, meaning that it is achieving high detection and low false-positive rates, it is ready to be used in the fraud detection workflow of your application.

Learn more

To fully understand models, make sure you also go through the following concepts:

- [Model training](#)
- [Model evaluation](#)
- [Feature importance](#)
- [White-box explanations](#)
- [Hyperparameter tuning](#)
- [AutoML](#)